

Glass Box Trading

AI proposes; deterministic gates dispose.

Thesis, and who it is for

An autonomous options trading agent whose every candidate, veto, and executed trade is public and broker-reconciled, because a system that manages risk unattended must be auditable by someone other than its builder. It is for traders and allocators who want unattended execution but distrust opaque "AI trading bot" claims: each decision is journaled before the agent acts, so a reviewer can check reasoning against outcome without taking anyone's word for it.

How it works

A scheduled cycle asks an LLM analyst (Alpaca MCP server, read-only market data) for defined-risk options candidates; the output is schema-validated, whitelist-constrained, and never touches an order. A pure, tested decision core scores every candidate against a fixed gate catalog — sleeve budgets, max loss, concentration, liquidity, session, lifecycle — and returns approve or veto with reasons. Only core-approved actions reach the executor, which places paper orders on the \$100k competition account and journals intent before submission, outcome after. That journal is append-only and public; the dashboard needs no login.

The four judging criteria, mapped to evidence

P&L: account PA376WIK2ATL, revision sha256:7b82959a344a7c7e, every fill linked to its intent and outcome. **Technology:** one end-to-end cycle in the video; the analyst's candidates, every gate verdict and every broker order id in the journal; core source and tests public. **Originality:** vetoes and no-trade cycles shown at the fidelity of executions. **Presentation:** demo at <https://glass-box-trading.vercel.app>, video, deck, preflight record.

Result at the presentation cutoff (2026-09-03T20:00:14.787Z)

Start equity \$100,000.00; equity at cutoff \$100,583.59; P&L \$583.59 (0.58%); realized \$586.00; unrealized \$0.00; unattributed -\$2.41 — a fee-shaped residual the journal cannot explain, so it is displayed and never assigned to a sleeve. Sleeve attribution covers realized P&L only: income \$230.00 over 5 filled spreads, convex \$356.00 from 1 filled call of 4 attempted. Concentration: one 4x QQQ long option that caught an overnight gap made \$356.00, 61% of the result. Peak simultaneous defined worst case \$3,421.00 (3.42% of the account), carried overnight; the measured max drawdown of \$33.15 comes from cycle-spaced samples across 2 sessions and does not sample the overnight move. The book is flat: zero positions, zero open orders. All figures are broker-derived.

Limitations

Two sessions of paper trading cannot prove edge, and no result here is evidence of alpha. Trading is paper only, never live; the sleeve split is declared, not backtested.

Two defects surfaced during the competition, both in `DECISIONS.md`. On the flatten day the runner could not price the three structures expiring the next session — its quote window starts at `EXPIRY_MIN_SESSIONS` — and the refusal reached only a discarded cycle report, so those closes were never submitted. The owner stood the writer down; the certified dead-man watchdog took over, journaled `HALT_WATCHDOG_TAKEOVER` and closed all three structures, filled within a second. Second defect: that watchdog's wrapper had killed the CLI on its first stderr line since arming, leaving the dead-man inert for a day; it was fixed at 17:41 CEST, an hour before the takeover. The safety net held when it was needed; neither defect is edited out.

Known API limitation. Alpaca has no conditional submit, so the gap between the pre-submit broker re-fetch and broker acceptance stays open; that re-fetch's completion is the declared linearization point. Manual account changes outside a durable halt are prohibited, and one that lands anyway is caught next cycle as `RESIDUE` or `HUMAN_ACTION`.